CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Software engineer for AI agents (MSc in Artificial Intelligence, 3 years of experience), most recently in the German energy sector. I build LLM agent systems, the Model Context Protocol (MCP) server that gives them tools and context, and the evaluation that shows whether their non-deterministic output can be trusted (precision/recall 33%/45% → 93%/90% against SAP-system data). I work directly with domain experts and business stakeholders to turn their processes into software.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer of an LLM-based agent system for the German energy sector, built end-to-end in Python (data ingestion, chunking, prompt engineering, indexing with agentic, deterministic, and monolithic strategies); worked with SAP domain experts on extracting knowledge from regulatory and technical documents, reporting directly to the CTO.
- Built a Model Context Protocol (MCP) server that exposes the knowledge graph as a tool with millisecond-latency retrieval, integrated into Claude Code for agentic development workflows.
- Built multi-agent extraction with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents in parallel) and evaluated its non-deterministic output against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Developed Expert Interface, a web app (React frontend, Python microservices backend) on Azure where domain experts review and approve the agents' knowledge-graph changes, keeping a human in the loop.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, Claude Code, RAG, GraphRAG, FastAPI, React, Neo4j, MongoDB, Postgres, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Advised business stakeholders on how agentic AI could support their decision-making workflows and translated their needs into a solution approach.
- Designed and deployed an agentic AI application with LangGraph that routes each request to vector database queries, general inference, or summarization.
- Built the full stack (React frontend, Python/FastAPI REST backend, API testing with Postman), containerized it with Docker, deployed it to AWS with CDK, and version-controlled it in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Built automated evals for a RAG assistant (generation and retrieval) with RAGAS and PytrecEval, comparing its answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Built event-driven AWS infrastructure (Lambda, SQS, DynamoDB) with CDK for PrismaChat RAG evaluation.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data.
- Built a Streamlit dashboard letting chemists simulate parameter changes and see predicted outcomes before running physical batches.

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.

PROJECTS

Project: Energy Customer Service Agent | TypeScript | MCP | Python | LLM APIs

- TypeScript customer service agent for an energy supplier that explains a bill and accepts a meter reading through MCP tools on a Python backend, with authentication, per-customer permissions, input validation, and a human handoff that passes the full conversation context.
- Client-agnostic journeys configured per client and shipped behind feature toggles, with guardrails on every tool call; an eval set of labelled conversations tests correctness and handoff decisions in CI, and tool calls are logged for observability and monitoring, with documentation and a runbook.

TECHNICAL SKILLSET

- **AI Agents:** autonomous and multi-agent systems, agent frameworks (LangGraph, LangChain), Model Context Protocol (MCP), LLM APIs (LiteLLM), prompt engineering, RAG, GraphRAG, evals for non-deterministic output (RAGAS, PytrecEval), Claude Code
- **Software Engineering:** Python, SQL, FastAPI, REST APIs, React (beginner), Git, GitLab CI/CD, Postman, Docker, Kubernetes
- **Cloud & Data:** AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3), Azure, Postgres, MongoDB, Neo4j, vector databases
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time-series analysis

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)